ME218: Solid Mechanics Lab

Stress Concentration in Perforated Plates

Instructor:
Prof. Sripriya Ramamoorthy

Group:
G08

Group members

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Abstract

In this project, we investigated stress concentration effects in plates with geometric discontinuities using both experimental and computational approaches. Epoxy specimens with different geometries, including square holes, elliptical holes, and filleted square holes, were fabricated to study how shape influences stress distribution. Photoelasticity was the primary experimental technique used to visualize stress fields, observe fringe patterns, and identify regions of high stress concentration.

To complement the experimental study, finite element analysis (FEA) simulations were carried out in ANSYS for the same geometries. The simulation results were compared with the experimental observations to validate the accuracy of the numerical models. A good agreement between the two approaches confirmed the reliability of the simulations.

Overview

This project aimed to analyze stress distribution and concentration in structural components using photoelasticity and validate the results through Finite Element Analysis (FEA). Understanding how complex stress patterns develop under loading is critical for improving design reliability and preventing structural failure, and traditional analytical solutions are often insufficient for irregular geometries, making experimental and computational approaches necessary.

The project was executed systematically, specimen geometries were designed, TPU moulds were 3D-printed for precision, and epoxy resin was cast to produce transparent photoelastic specimens. Testing was carried out using a polariscope under controlled loading, while simultaneous ANSYS simulations with appropriate material properties and boundary conditions were developed for the same geometries.

Results showed a high degree of agreement between the photoelastic fringe patterns and FEA stress contours, particularly in identifying stress concentration regions. Minor discrepancies were attributed to air bubbles, residual stresses, and material model simplifications. Future work could involve improved material characterization, dynamic loading conditions, more complex geometries, and automation of fringe pattern analysis through image processing.

Name	Contributions
Tanushree	Specimen design, manufacturing, data acquisition, documentation
Durva	FEA simulations, manufacturing, data acquisition, documentation
Tanishk	Specimen design, data acquisition, documentation, material sourcing
Vidit	Specimen design, manufacturing, FEA simulations, documentation

Table 1: Group Member Contributions

Materials And Methods

Materials

The primary materials used in this study were epoxy resin and Thermoplastic Polyurethane (TPU) moulds. The epoxy resin was selected due to its transparency and suitability for photoelastic analysis, enabling clear visualization of stress patterns. It was procured from a local hardware store Mangaldeep to ensure quick availability. The TPU moulds were fabricated using in-house 3D printing facilities available within the institute, allowing precise control over geometry and repeatability of specimens. The flexibility and durability of TPU made it suitable for mould fabrication and removing the hardened resin.

Methods

The methodology involved both experimental and computational techniques to study stress distribution. Photoelasticity was employed as the primary experimental method, where epoxy resin specimens were subjected to loading and observed under polarized light. The resulting fringe patterns provided insight into stress concentrations and distribution within the material. This method enabled qualitative as well as semi-quantitative analysis of internal stresses.

In addition to experimental methods, Finite Element Analysis (FEA) simulations were performed to validate and complement the observations. The simulations were carried out by creating accurate geometric models and assigning appropriate material properties and boundary conditions. The results from FEA were compared with experimental findings to ensure consistency and reliability of the analysis.

Testing Equipment and Lab Details

All experimental tests were conducted using a standard photoelasticity setup available in the Solid Mechanics Laboratory, Department of Mechanical Engineering. The setup consisted of a polariscope arrangement used to visualize isochromatic fringe patterns under controlled loading conditions. Computational simulations were performed using ANSYS software on personal laptops, providing detailed stress distribution results for comparison with experimental data.

Item Name	**Quantity**	**Cost (Rs.)**
Bondtite epoxy resin	1 pc	1008
Araldite epoxy resin	1 pc	550
Total		**Rs. 1558**

Table 2: Consumables used

Fillet v/s Non-Fillet Edges

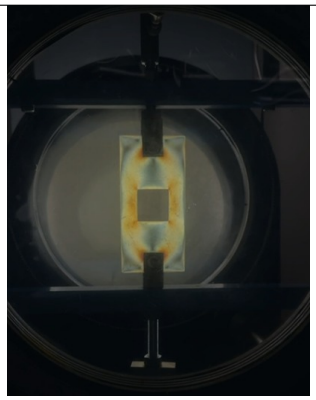
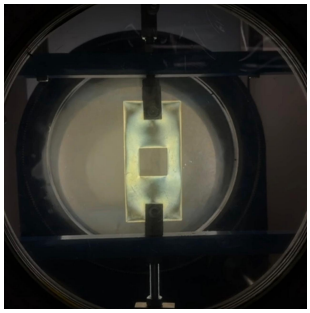
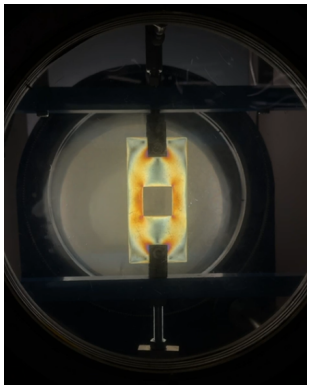
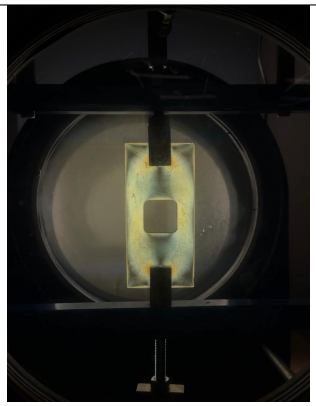
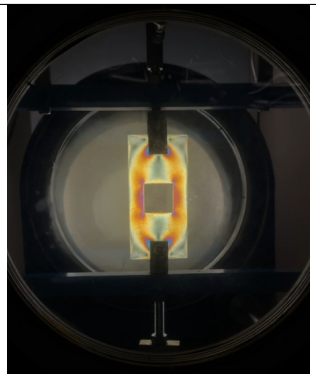
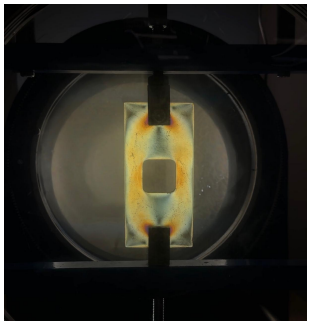
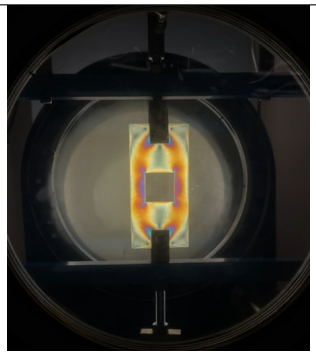
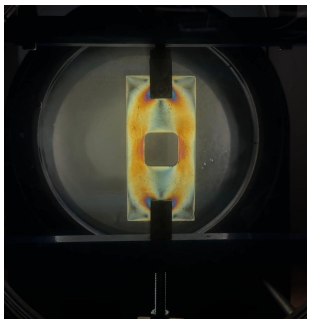
Load (N)	Without Fillet	With Fillet
18.5		
23		
26		
26.7		

Table 3: Stress concentration comparison (Part 1)

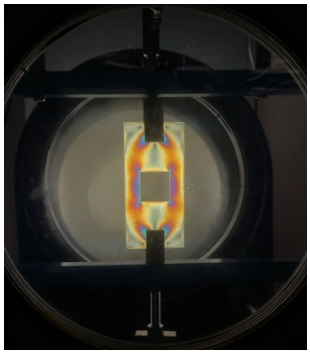
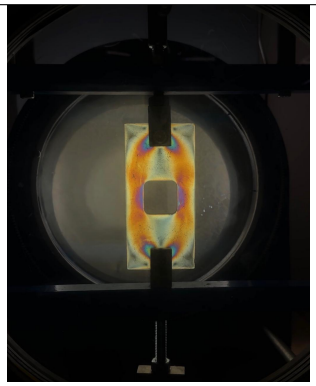
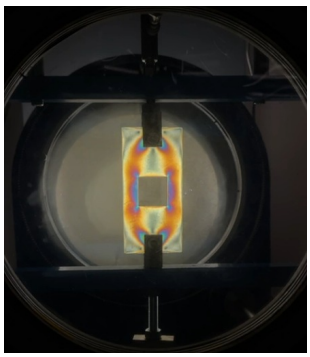
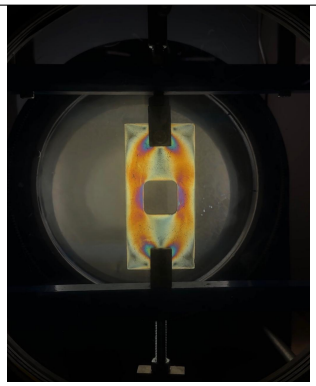
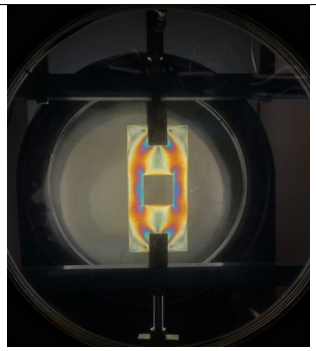
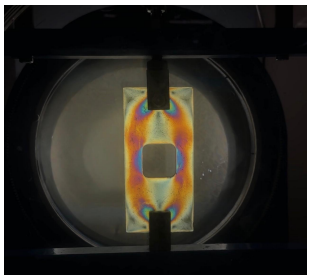
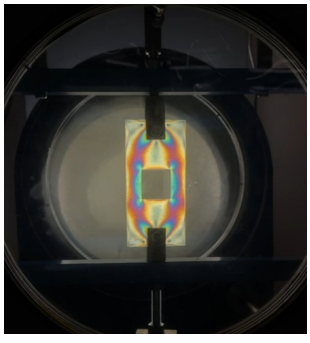
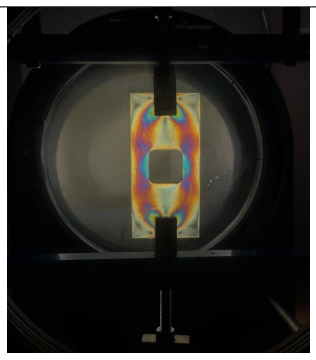
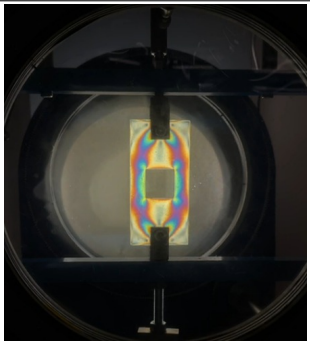
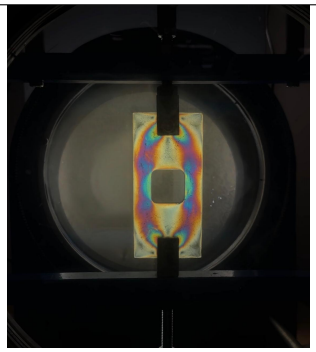
Load (N)	Without Fillet	With Fillet
29		
30		
33		
34		
35		

Table 4: Stress concentration comparison (Part 2)

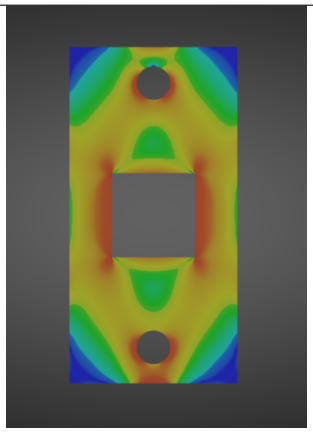
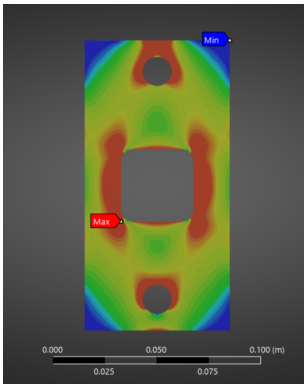
	Without Fillet	With Fillet
Simulation at Load = 20N		

Table 5: Stress concentration comparison (Part 3)

Observations

Sample thickness $t = 15\,mm = 0.015\,m$

Sample width $w = 40\,mm$

Fillet

Fringe orders (from pattern estimation):

$$N_a = 0.5(0.9) + 0.5(0.8) = 0.85$$

$$N_b = 0.35(0.45) + 0.75(0.6) = 0.6075$$

Using $\sigma = \dfrac{N\,F}{t}$

$$SCF_{fillet} = \frac{N_a}{N_b} = \frac{0.85}{0.6075} \approx 1.4$$

Non Fillet

Fringe orders (from pattern estimation):

$$N_a = 0.4(1.22) + 0.6(0.9) = 1.028$$

$$N_b = 0.35(0.45) + 0.75(0.6) = 0.6075$$

$$SCF_{non-fillet} = \frac{N_a}{N_b} = \frac{1.028}{0.6075} \approx 1.7$$

Eliptical Hole

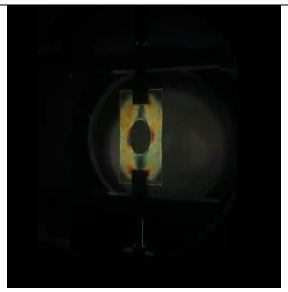
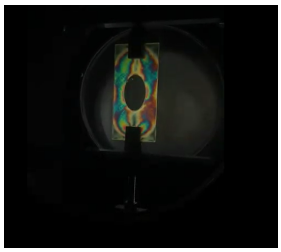
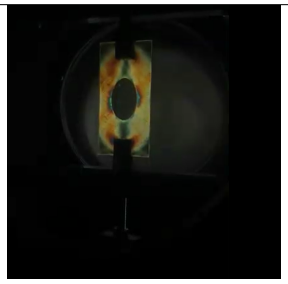
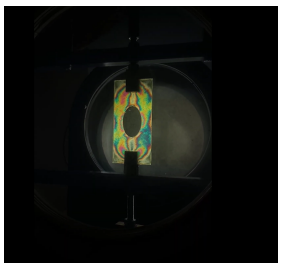
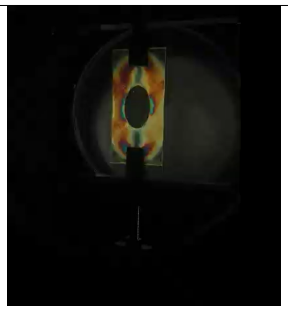
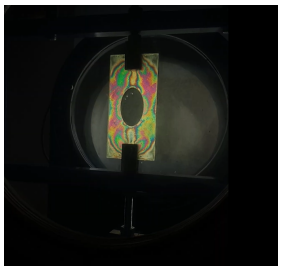
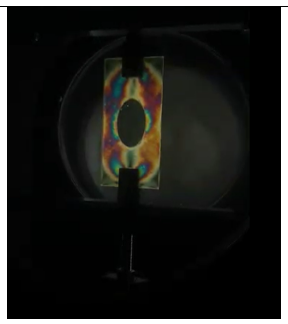
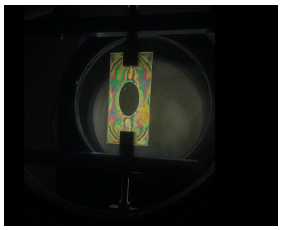
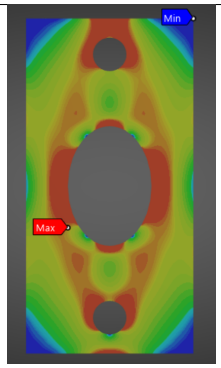
Load (N)	Stress	Load (N)	Stress
2.5		5.5	
3		7	
4.5		8	
5		10	
Simulation @ Load = 20N			

Table 6: Stress concentration

Elliptical Hole

Fringe orders (from pattern estimation):

$$N_a = 0.4(1.22) + 0.4(0.9) + 0.2(0.8) = 1.008$$

$$N_b = 0.3(0.45) + 0.5(0.9) + 0.2(0.6) = 0.885$$

$$SCF_{elliptical} = \frac{N_a}{N_b} = \frac{1.008}{0.885} \approx 1.14$$

Conclusion

- Photoelasticity was successfully demonstrated as an effective experimental technique for visualizing and analyzing stress concentration in epoxy specimens with varying geometric discontinuities.
- Comparing square hole specimens with and without fillets, and an elliptical hole geometry, clearly established the influence of geometric design on stress distribution.
- Experimentally determined SCF values of 1.4 (fillet) and 1.7 (non-fillet) confirmed that introducing fillets at sharp corners significantly reduces local stress concentrations.
- The elliptical hole yielded the lowest SCF of approximately 1.14, underscoring the advantage of smooth, curved boundaries in minimizing stress risers.
- FEA simulations in ANSYS showed strong qualitative agreement with photoelastic fringe patterns, particularly in identifying locations and relative magnitudes of stress concentrations.
- Minor discrepancies were attributed to residual stresses, fabrication-induced air bubbles, and simplifying assumptions in material models.
- The project validated the complementary nature of experimental and computational methods in stress analysis, reinforcing that sharp geometric discontinuities should be avoided to enhance structural integrity and fatigue life.